

🌐 Challenge: Deploy a Lightweight Web App to EKS (Configurable Color)

This challenge invites you to **create a lightweight web application** using your preferred tech stack (Python/Flask, Java/Spring Boot, Node.js/Express, C#/ASP.NET, etc.), containerize it, and deploy it to your existing **EKS cluster**. The app should display a simple message with **configurable color**, controlled via environment variables.

🎯 Objective

1. Create a minimal web app that displays:

```
```
Hello World!!
```
```

Styled in **blue** initially.

2. The message color must be **controlled via environment variable**, e.g.:

```
```
TEXT_COLOR=red
```
```

3. Containerize your application using Docker.

4. Deploy it as a **Pod** to your existing **EKS cluster**.

5. Update the color by **changing the environment variable** in your Pod definition and re-deploy.

🛠️ Choose Your Language / Framework

You may implement this in any of the following (or your own preferred) stacks:

Stack	Example Framework
Python	Flask
Java	Spring Boot
Node.js	Express
C#	ASP.NET Core
Go	net/http
Rust	Rocket / Actix

📦 Requirements

- Your app must:
 - Accept a `TEXT_COLOR`` environment variable

- Render `Hello World!!` with the color specified
- Run on port `5000` or `8080`
- Be packaged using Docker

🚀 Kubernetes Deployment

- Write a `pod.yaml` manifest
- Inject the environment variable like this:

```
```yaml
env:
 - name: TEXT_COLOR
 value: red
```
```

```
- Deploy with:
```bash
kubectl apply -f pod.yaml
```
```

```
- (Optional) Expose via LoadBalancer or use:
```bash
kubectl port-forward pod/<your-pod> 5000:5000
```
```

✅ Deliverables

- `app.(py|java|js|cs|go|...)`
- `Dockerfile`
- `pod.yaml`
- Screenshot with initial (blue) and updated (red) output

🧠 Hints

- For Flask: <https://flask.palletsprojects.com/en/2.3.x/quickstart/>
- For Spring Boot: <https://spring.io/guides/gs/spring-boot/>
- For Express.js: <https://expressjs.com/en/starter/hello-world.html>
- For ASP.NET: <https://learn.microsoft.com/en-us/aspnet/core/tutorials/first-web-api>
- For Docker: <https://docs.docker.com/language>

🎁 Bonus

- Push image to ECR
- Use ConfigMap to inject env vars
- Package using a Helm chart

📌 Reminder

Make sure your app **uses the environment variable dynamically** and not hardcoded color values!

Happy learning 🚀